STAT 314500 - Applied Partial Differential Equations
Assignment 1: Heat equation

This is an individual assignment. You may collaborate with others but are expected to submit your own work. Show all relevant work. Incomplete work will be docked points.

Note: if a question is marked with an asterisk, *, and appears vague - that's on purpose. Questions marked * are conceptual short-answer questions. Do your best to reason these through, and describe that reasoning concisely. We are looking more for reasoning than precision.

Reading: Chapter 1 in Applied Partial Differential Equations with Fourier Series and Boundary Value Problems (Richard Haberman).

Question 1

Consider the one dimensional heat equation

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}, \quad x \in (0, L), t > 0$$

with initial condition

$$u(x, 0) = f(x) .$$

Find the equilibrium/steady state temperature distribution ($\lim_{t\to\infty} u(x, t)$) when the boundary conditions are of the following type:

1. Dirichlet type (5 points)

$$u(0, t) = T_1, \quad u(L, t) = T_2 .$$

2. Neumann type (5 points)

$$\frac{\partial u}{\partial x}(0, t) = 0, \quad \frac{\partial u}{\partial x}(L, t) = 0 .$$

3. Robin type (5 points)

$$u(0, t) = T, \quad \frac{\partial u}{\partial x}(L, t) + u(L, T) = 0 .$$

Question 2

Consider the one dimensional heat equation with a constant source $Q \neq 0$

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} + Q, \quad x \in (0, L), t > 0$$

with initial condition

$$u(x, 0) = f(x)$$

and boundary conditions

$$\frac{\partial u}{\partial x}(0, t) = 0, \quad \frac{\partial u}{\partial x}(L, t) = 0 .$$

1. Show that there does not exist any equilibrium temperature distribution. Briefly explain why this is true physically. (5 points)

2. Calculate the thermal energy in the entire rod. (5 points)

Question 3 Suppose

$$\begin{cases} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} & = 4, \quad x \in (0, L), t > 0 \\ u(x,0) & = f(x) \\ \frac{\partial u}{\partial x}(0,t) & = 5 \\ \frac{\partial u}{\partial x}(L,t) & = 6 \end{cases}$$

Calculate the total thermal energy in the rod as a function of time. (5 points)

Question 4 Consider the polar coordinate representation $(x, y) = (r\cos\theta, r\sin\theta)$.

1. Using $r^2 = x^2 + y^2$, show that (4 points)

$$\frac{\partial r}{\partial x} = \cos\theta, \quad \frac{\partial r}{\partial y} = \sin\theta, \quad \frac{\partial \theta}{\partial x} = -\frac{\sin\theta}{r}, \quad \frac{\partial \theta}{\partial y} = \frac{\cos\theta}{r}$$

2. Show that $\hat{r} = \cos\theta\hat{\imath} + \sin\theta\hat{\jmath}, \hat{\theta} = -\sin\theta\hat{\imath} + \cos\theta\hat{\jmath}$. (2 points)

3. Using chain rule, show that $\nabla = \hat{r}\frac{\partial}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial}{\partial \theta}$. (4 points)

Question 5

Assume that the temperature is circularly symmetric, i.e, $u = u(r,t)$, where $r^2 = x^2 + y^2$. The goal of this problem is to derive the heat equation with circular symmetric geometry without sources.

1. Consider any circular annulus $a \le r \le b$. Show that the total heat energy is $2\pi \int_a^b c\rho u r \mathrm{d}r$. ($c$ is the specific heat and ρ is the mass density) (3 points)

2. Show that the flow of heat energy per unit time out of the annulus at $r = b$ is

$$-2\pi b K_0 \frac{\partial u}{\partial r}|_{r=b}.$$

   A similar result hold for $r = a$. (K_0 is the thermal conductivity). (3 points)

3. Using the first and second parts, show that (4 points)

$$\frac{\partial u}{\partial t} = \frac{k}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right).$$